生物统计和生物信息引论

Lecture 4

方差分析（ANOVA）

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Reading: Variance component (Chapter 1 – 3)

By Charles E. McCulloch, George Casella, and Shayle R. Searle

Tip: If you can read Chapters 4–6 independently, please contact me about a research project.

引导案例 I：番茄品种产量实验

Consider a home gardener carrying out a small experiment

24 tomato plants, **6 plants** of each of **4 varieties**

Comparison of the four varieties in the $12' \times 8'$ garden space. Each plant is allocated randomly to one of the $2' \times 2'$ squares.

If y_{ij} is the yield of fruit from plant j of variety i, a possible model for y_{ij} would be

$$E(y_{ij}) = \mu_i$$



多组比较

The Goal: We want to know if the mean yield $\mu_i, i = 1, \ldots, r$ differs across these 4 specific ($r = 4$) varieties.

We must compare every possible pair:

(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), and (3, 4)

This results in 6 separate pairwise comparisons.

Null Hypothesis (H_0): $\mu_1 = \mu_2$

What do we need for t-tests?
What's the sample size?

$$t = \frac{\bar{Y}_1 - \bar{Y}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Potential Limitations

$$t = \frac{\bar{Y}_1 - \bar{Y}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Type I Error Inflation: If we set if $\alpha = 0.05$ for each of the 6 tests, the probability of making at least one false discovery (concluding a variety is better when it isn't) increases significantly. For 6 comparisons, the overall error rate inflates significantly ($1 - 0.95^6 \approx 26.5\%$).

Inefficiency: Each t-test only uses 12 plants (6 from each of 2 varieties) to estimate variance.

- Everyone is lazy: Move beyond simple one-to-one comparisons to look at the entire experimental design at once, protecting our error rate and maximizing our data's power.

- We want to use all 24 plants simultaneously to estimate the “background noise”, providing a much more stable and accurate baseline for comparison —— ANOVA

方差分析中常用概念

- **Factor:** The categorical variable being investigated to see if it affects the outcome. (tomato variety / clinics)

- **Response (Y):** The quantitative outcome that is measured and compared in the experiment. (yield / measured response)

- **Level (r):** each variety is called a level of the factor. ($r = 4$ varieties / 15 different clinics)

- **Experimental units:** The independent objects or individuals that receive a specific treatment.(Each plant or square / Patient)

单因素ANOVA模型

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, \ldots r, j = 1, \ldots, n_i.$$

- μ_i is the theoretical mean of all observations at level i
- ε_{ij} *are* iid $N(0, \sigma^2)$, implies the Y_{ij}'s are independent $N(\mu_i, \sigma^2)$
- n_i: number of cases for *i*-th factor, $n_T = \sum_{i=1}^{r} n_i$
- We call it cell-means model

Question: Does μ_i depend on i?

$H_0: \mu_1 = \mu_2 = \cdots = \mu_r = \mu$ No group difference

H_a : at least one μ_i different

固定效应模型 (Fixed Effects Model)

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, \dots r, j = 1, \dots, n_i.$$

- Assumption: μ_i are fixed, unknown constants chosen specifically by the researcher.
- Parameters ($r + 1$ total):
  - The specific group means: $\mu_1, \mu_2, \dots, \mu_r$.
  - The error variance: σ^2.

For fixed effect model, we are estimating specific means rather than a distribution of means. The parameters of this model are the specific group means and the shared background noise or variance.

- **Independence**: Each data point must be completely independent of the others.
- **Normality**: The data within each group (or the model's residuals) should follow a normal distribution.
- **Homoscedasticity**: The variance (spread of data) should be approximately equal across all compared groups.

方差分解思想 (Variance Decomposition)

Although ANOVA is about “Variance”, its goal is to infer whether population means are equal.

It achieves this by analyzing the **sources of variation** in the observed data.

总变异 (Total Variation)

↓

组间变异 (Between-Group)

由因子水平（如番茄品种）引起

+

组内变异 (Within-Group)

由随机误差或未受控因素引起

Unequal group mean -> Between-Group variation

方差分解思想 (Variance Decomposition)

For the j-th observation in the i-th group, Y_{ij}, the deviation from the grand mean can be decomposed as:

$$Y_{ij} - \bar{Y}_{\cdot\cdot} = Y_{ij} - \bar{Y}_{i\cdot} + \bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}$$

$$\bar{Y}_{\cdot\cdot} = \sum\sum Y_{ij}/n_T$$

$$\bar{Y}_{i\cdot} = \sum Y_{ij}/n_i$$

Total Deviation:

Individual deviation from the grand mean.

Error Deviation:

Individual deviation from group mean

(Residual/Random Error)

Treatment Deviation:

Group mean deviation from grand mean

(Factor Effect)

Squaring both sides and summing over all i, j (cross-product terms are zero) yields:

$$\sum_i \sum_j (Y_{ij} - \bar{Y}_{\cdot\cdot})^2 = \sum_i n_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2$$

$$SST = SSR + SSE$$

Total Sum of Squares

Total variation

Regression/Treatment Sum of Squares

Between-group sum of squares
Reflects factor level effects

Error Sum of Squares

Within-group sum of squares
Reflects random error

方差分解思想 (Variance Decomposition)

$$Y_{ij} - \bar{Y}_{\cdot\cdot} = (Y_{ij} - \bar{Y}_{i\cdot}) + (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}) \qquad (Y_{ij} - \bar{Y}_{\cdot\cdot})^2 = \left[(Y_{ij} - \bar{Y}_{i\cdot}) + (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})\right]^2$$

$$\begin{aligned}
\sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{\cdot\cdot})^2 &= \sum_{i=1}^{r}\sum_{j=1}^{n_i}\left[(Y_{ij} - \bar{Y}_{i\cdot}) + (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})\right]^2 \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})^2 + \sum_{i=1}^{r}\sum_{j=1}^{n_i}(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 + 2\sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}) \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})^2 + \sum_{i=1}^{r} n_i(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 + 2\sum_{i=1}^{r}\left((\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})\right) \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})^2 + \sum_{i=1}^{r} n_i(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 + 2\sum_{i=1}^{r}(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})\cdot 0 \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i\cdot})^2 + \sum_{i=1}^{r} n_i(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 \\
&= SSE + SSR
\end{aligned}$$

固定效应模型 (Fixed Effects Model)

Estimate μ_i by the sample mean of the observations at level i

$$\hat{\mu}_i = \bar{Y}_{i\cdot}$$

$$\bar{Y}_{\cdot\cdot} = \sum\sum Y_{ij}/n_T$$

$$\bar{Y}_{i\cdot} = \sum Y_{ij}/n_i$$

Unbiasedness

$$E[\hat{\mu}_i] = E[\bar{Y}_{i\cdot}] = E\left[\frac{1}{n_i}\sum_{j=1}^{n_i} Y_{ij}\right] = \frac{1}{n_i}\sum_{j=1}^{n_i} E[Y_{ij}] = \frac{1}{n_i}(n_i\mu_i) = \mu_i$$

For each level i, also estimate the variance

$$s_i^2 = \sum (Y_{ij} - \bar{Y}_{i\cdot})^2/(n_i - 1)$$

s_i^2 represents the within-group variance for level i. It provides a measure of how "noisy" the data is within that specific category.

Unbiasedness (Bessel's Correction): Dividing by n_i creates a **biased** estimator that systematically underestimates σ^2. Using $n_i - 1$ ensures the estimator is **unbiased**: $E[s_i^2] = \sigma^2$.

Noise variance estimation

s_i^2 in each group are combined to estimate σ^2

$$s_i^2 = \sum (Y_{ij} - \bar{Y}_{i\cdot})^2/(n_i - 1)$$

If n_i were equal, could compute s^2 by averaging s_i^2's

More general formula, weighted sum (proportional to sample size) of each s_i^2

$$s^2 = \frac{\sum (n_i - 1)s_i^2}{\sum (n_i - 1)}$$

$$= \frac{\sum (n_i - 1)s_i^2}{n_T - r}$$

where n_T is the total number of observations.

固定效应模型 (Fixed Effects Model)

ANOVA Table

Source of Variation	df	SS	MS
Model	$r-1$	$\mathrm{SSR}=\sum n_i(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2$	$\mathrm{MSR}=\frac{\mathrm{SSR}}{r-1}$
Error	n_T-r	$\mathrm{SSE}=\sum\sum(Y_{ij}-\bar{Y}_{i\cdot})^2$	$\mathrm{MSE}=\frac{\mathrm{SSE}}{n_T-r}$
Total	n_T-1	$\mathrm{SST}=\sum\sum(Y_{ij}-\bar{Y}_{\cdot\cdot})^2$	

Both Sum of squares and degrees of freedom sum up to the totals

Expected Mean Squares:

All means squares are random variables; and SSE is independent of SSR.

Please show

$$E(MSE)=\sigma^2$$

$$E(MSR)=\sigma^2+\frac{\sum n_i(\mu_i-\mu_\cdot)^2}{r-1},\ \mu_\cdot=\frac{\sum n_i\mu_i}{n_T}$$

Hypothesis testing

Since the factor levels only affect the cell means:

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, \ldots, r, \qquad j = 1, \ldots, n_i.$$

The goal is to test

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_r = \mu$$

H_a : at least one μ_i different

Intuition: If the null hypothesis is true, $Var(Y_{ij}) = Var(\epsilon_{ij})$

固定效应模型 (Fixed Effects Model)

F - test

If H_0 is true, MSR is unbiased estimate of σ^2。

The test statistic is defined as

$$F^* = \frac{MSR}{MSE} \sim F(r-1, n_T - r)$$

When $r = 2$, $F = t^2(n_T - 1)$.

Intuitive Interpretation: The F-value is the "Signal-to-Noise" ratio. A larger F-value indicates that between-group variation is significant relative to background noise, favoring rejection of H_0.

Let $U \sim \chi^2(d_1)$ and $V \sim \chi^2(d_2)$ be independent. Then:

$$F = \frac{U/d_1}{V/d_2} \sim F(d_1, d_2)$$

$$\frac{SSE}{\sigma^2} = \sum\sum \frac{(Y_{ij} - \bar{Y}_{i.})^2}{\sigma^2} \sim \chi^2(n_T - r)$$

$$\frac{SSR}{\sigma^2} = \sum n_i \frac{(\bar{Y}_{i.} - \bar{Y}_{..})^2}{\sigma^2} \sim \chi^2(r-1)$$

Under H_0

$$F^* = \frac{\frac{SSR/\sigma^2}{r-1}}{\frac{SSE/\sigma^2}{n_T - r}} = \frac{MSR}{MSE}$$

ANOVA 与回归的联系

ANOVA can be regarded as regression model

$$Y = X\beta + \epsilon$$

Y: The vector of observations ($n_T \times 1$).

X: The **Design Matrix** ($n_T \times r$), which encodes group membership.

β: The parameter vector. In the **Cell Means Model**, $\beta = [\mu_1, \mu_2, \ldots, \mu_r]^T$.

ϵ: Random error, assuming $\epsilon \sim N(0, \sigma^2 I)$.

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{n_r} \end{pmatrix}_{n_T \times r}$$

Balanced Design: in example 1,
each column contains exactly 6 ones,
indicating a balanced sample size $n_i = 6$

$$X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{24\times 4}$$

Variety 1 (6 rows)
Variety 2 (6 rows)
Variety 3 (6 rows)
Variety 4 (6 rows)

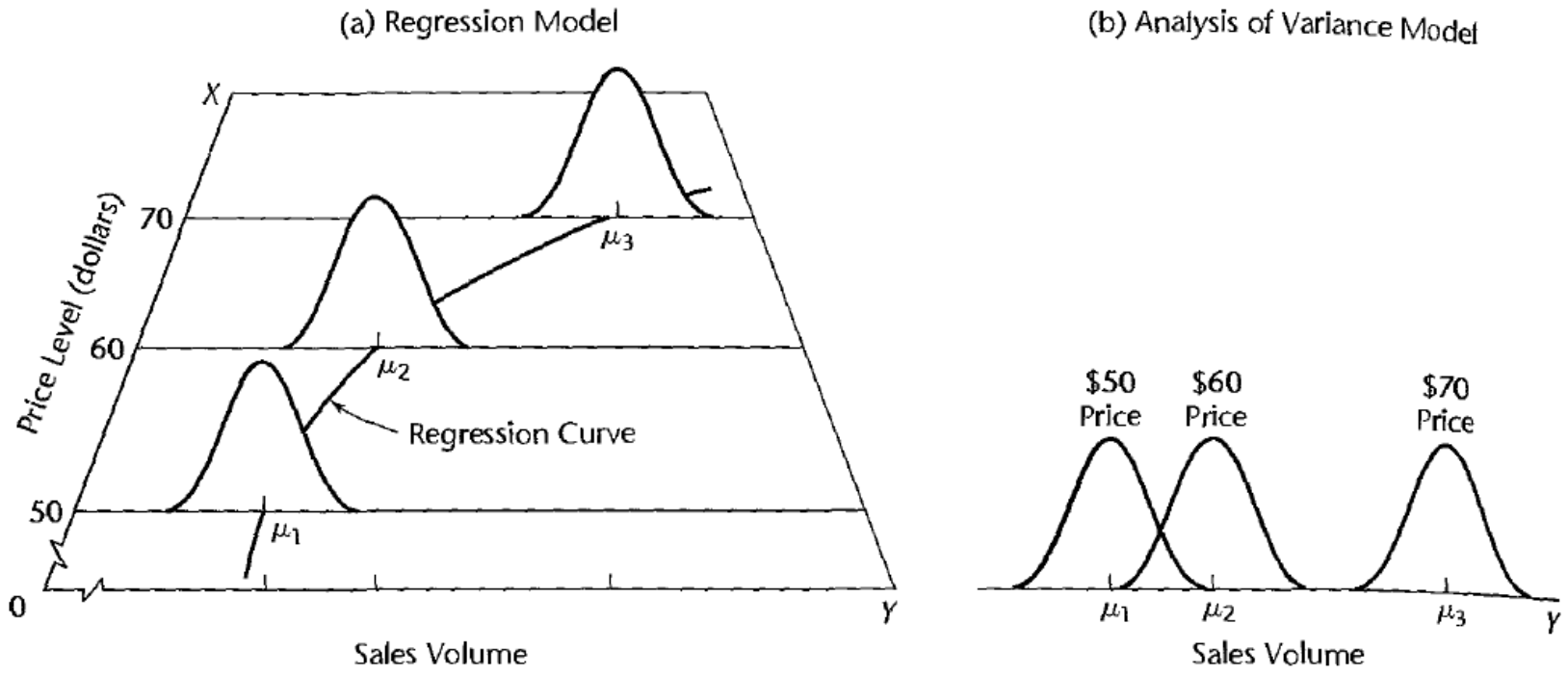
Unbalanced Design:

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$X_{ij} = 1$ if the i-th plant belongs to variety j

ANOVA 与回归的联系

FIGURE 16.1 Relation between Regression and Analysis of Variance Models.



Reference: Michael H. Kutner “Applied Linear Statistical Models(Fifth Edition)”

ANOVA 与回归的联系(3)

By calculating the expected value $E(Y) = X\beta$ we see how individual observations are mapped to their respective group means:

$$E[Y_{ij}] = \text{Row}_i X \cdot \beta = [0, \ldots, 1, \ldots, 0] \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_i \\ \vdots \\ \mu_r \end{pmatrix} = \mu_i \qquad X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{n_2} & \ldots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{1}_{n_r} \end{pmatrix}_{n_T \times r}$$

OLS solution:

$$X^\top X = \begin{pmatrix} n_1 & 0 & \cdots & 0 \\ 0 & n_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n_r \end{pmatrix} \qquad X^\top Y = \begin{pmatrix} \sum_{j=1}^{n_1} Y_{1j} \\ \sum_{j=1}^{n_2} Y_{2j} \\ \vdots \\ \sum_{j=1}^{n_r} Y_{rj} \end{pmatrix} \qquad \hat{\beta} = (X^\top X)^{-1} X^\top Y = \begin{pmatrix} \frac{1}{n_1} \sum_{j=1}^{n_1} Y_{1j} \\ \frac{1}{n_2} \sum_{j=1}^{n_2} Y_{2j} \\ \vdots \\ \frac{1}{n_r} \sum_{j=1}^{n_r} Y_{rj} \end{pmatrix}$$

From Global to Pairwise Selection

Step 1: Global F-Test (ANOVA)

- Test the overall hypothesis $H_0: \mu_1 = \mu_2 = \cdots = \mu_r = \mu$
- This tells us if a difference exists, but not where.

Step 2: Pairwise Comparisons

- Only proceed if the Global F-test rejects H_0.
- Identify specific differences between groups (e.g., t test between Group 1 vs. Group 2).

Why is pairwise comparison difficult?

> Simultaneous testing: for each CI containing zero or the test statistic exceeding critical values H_0 concluded

In the example 1, we have 6 pairwise comparisons => 6 t-tests and pvalues

- Control family-wise error rate (FWER)

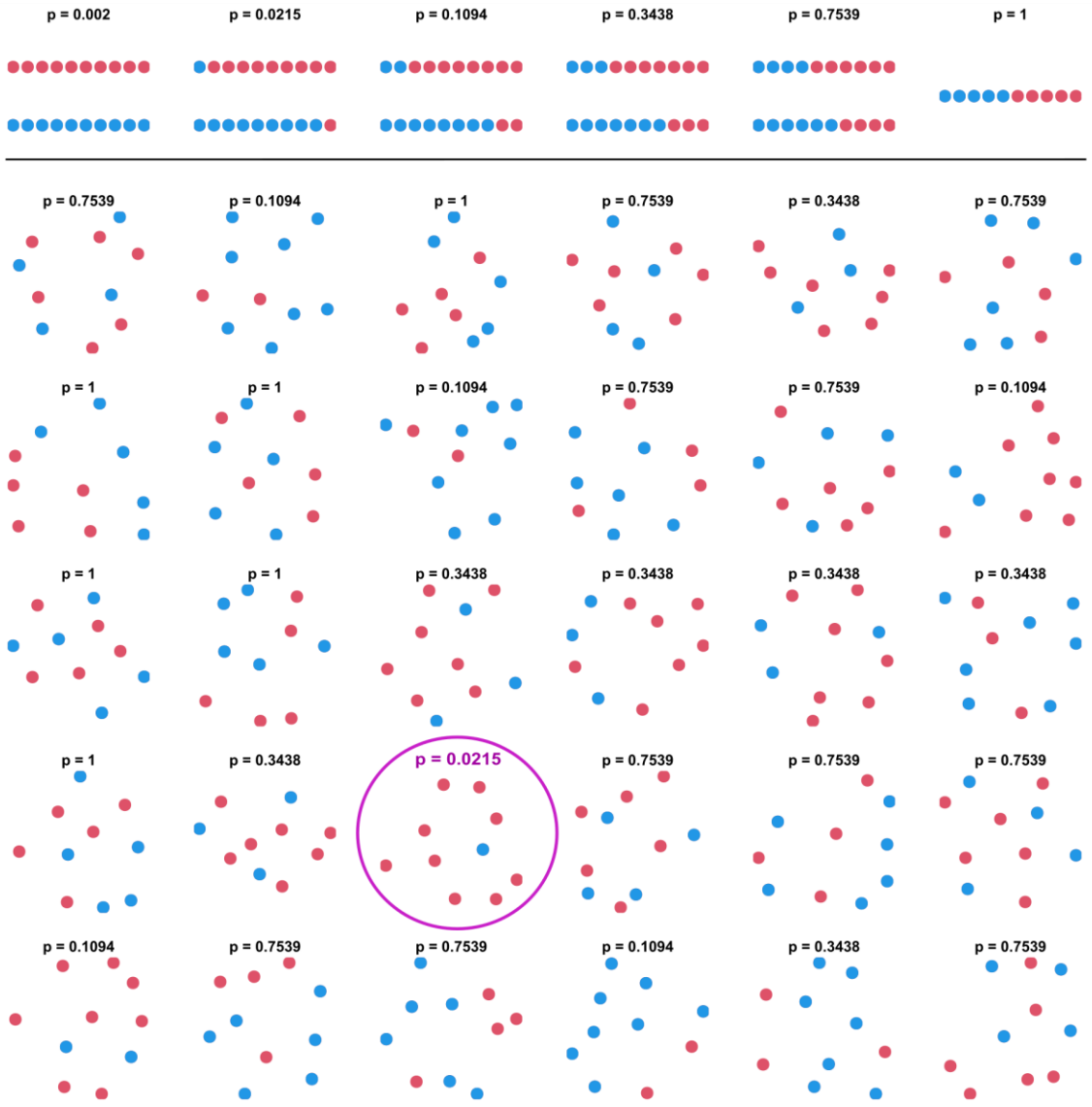
The probability of making at least one false positive (Type I error) when performing multiple hypothesis tests.

$$FWER = P(\text{at least one false positive}) = P(FD \geq 1)$$

As few false discoveries as possible.

Construct family-wise confidence interval

多重比较问题: Why it is difficult?



https://en.wikipedia.org/wiki/Multiple_comparisons_problem

- Experiment: 30 independent samples (10 dots each).
- Null Hypothesis H_0: $p(red) = p(blue) = 0.5$ (probability = 0.5).
- The Reality: ALL 30 samples were generated purely by random chance. H_0 is strictly true.

If you run enough tests, you are guaranteed to find a "significant" result just by random noise.

Without statistical correction, **p-values lose their protective meaning** when doing multiple comparisons.

Bonferroni Procedure for Identification

Principle

For n tests (t test/p-values in previous example) , to keep overall error at α, set significance level per test to:

$$\alpha_{\text{adj}} = \frac{\alpha}{n}$$

Example: n = 100, α = 0.05 -> New threshold: 0.0005

Critique: Very conservative.

- Usage: A finite number of pre-planned comparisons.
- does not rely on specific distribution assumptions, works for dependent tests.
- Cons: Extremely conservative when n is large, resulting in very low power.

https://en.wikipedia.org/wiki/Bonferroni_correction

Tukey Multiple Comparison Procedure

- Goal: compare all pairs of group means (e.g., A vs B, A vs C, B vs C) while controlling FWER

$$H_0: \ \mu_i = \mu_j \quad \text{for all } i \neq j$$

$$H_{0,ij}: \ \mu_i - \mu_j = 0$$

- Core: Based on the **Studentized Range Distribution** q.

Test statistic:

$$D_{ij} = \frac{|\bar{Y}_{i\cdot} - \bar{Y}_{j\cdot}|}{\sqrt{\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}$$

Studentized range distribution: $Y_1, ..., Y_r \overset{iid}{\sim} N(\mu, \sigma^2)$, estimate s^2 of σ^2 is based on ν degrees of freedom and independent from Y_i,

$$q(r, v) = \frac{\max(Y_i) - \min(Y_i)}{s}$$

Reject $H_{0,ij}$ if

$$D_{ij} > q_{1-\alpha}(r, \nu), \qquad \nu = N - r$$

Simultaneous $100(1-\alpha)\%$ confidence interval:

$$(\bar{Y}_{i\cdot} - \bar{Y}_{j\cdot}) \pm q_{1-\alpha}(r, \nu)\sqrt{\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

https://en.wikipedia.org/wiki/Tukey%27s_range_test

Scheffé Multiple Comparison Procedure

A general formulation : contrast, a linear combination of group means
$L = \sum_i c_i \mu_i$ with $\sum_{i=1}^{r} c_i = 0$

$$H_0 : L = 0$$

Simultaneous estimation for L: $\hat{L} = \sum c_i \bar{Y}_{i\cdot}$ and confidence intervals are in the form $\hat{L} \pm Ss\{\hat{L}\}$, where

$$S^2 = (r-1)F(1-\alpha; r-1, n_T - r),\ s^2\{\hat{L}\} = MSE \sum \frac{c_i^2}{n_i}$$

- Usage: All possible contrasts. Not just pairwise, but complex linear combinations (e.g., $\frac{\mu_1+\mu_2}{2} - \mu_3$).

$$L_1, L_2, L_3 \ldots.$$

https://en.wikipedia.org/wiki/Scheff%C3%A9%27s_method

多重比较三种方法

Comparison between Tukey, Scheffé and Bonferroni Procedure

$$\text{Tukey test:} \quad \frac{\sqrt{2}\,|\hat{D}|}{s\{\hat{D}\}} \le q(1-\alpha;\ r,\ n_T - r)$$

$$\text{Scheffé test:} \quad \frac{\hat{L}^2}{(r-1)\,s^2\{\hat{L}\}} \le F(1-\alpha;\ r-1,\ n_T - r)$$

$$\text{Bonferroni test:} \quad \frac{|\hat{L}|}{s\{\hat{L}\}} \le t\left(1 - \frac{\alpha}{2g};\ n_T - r\right)$$

所有 pairwise comparisons

任意线性 contrast

少量预先计划好的比较 （g: number of contrasts ）

引导案例 II：胰岛素临床诊所研究

Suppose a new form of injectable insulin is being tested using 15 different clinics in New York State. It is not unreasonable to think of those clinics as a **randomly chosen** sample of clinics from a **population of clinics** in New York. If clinic i has n_i patients in the trial and the measured response of patient j in clinic i is y_{ij} , a possible model

$$E(y_{ij}) = \mu + \tau_i, \text{for } j = 1, \dots, n_i$$

Note the underlying assumption is different from Example I. The τ_i is a clinic-specific effect.



固定效应vs随机效应

- Consider factor with numerous levels
- Want to draw inference on population of levels
- Not concerned with any specific levels
- Example of difference (1=fixed, 2=random)

1 Compare reading ability of 10 2nd grade classes in NY

2 Compare variability among all 2nd grade classes in NY

In problem 1, select a = 10 specific classes of interest, and randomly choose n students from each classroom

In problem 2, randomly choose n students from each classroom Level Chosen randomly, and inference on population

随机效应模型 (Random Effects Model)

Random Effects: The factor levels (e.g., specific clinics or personnel officers) are not fixed categories of interest, but are instead considered a random sample drawn from a much larger population.

$$Y_{ij} = \mu_. + \tau_i + \epsilon_{ij}$$

- $\mu_.$ is the "reference value", or the overall mean
- τ_i is the i-th treatment effect, unlike the fixed model, τ_i are random variables, not parameters.

$$\tau_i \sim N(0, \sigma_\tau^2)$$

- One may take $\tau_i = \mu_i - \mu$ from the cell mean model

Two modeling strategy

- Fixed Effects Model

$$Y_{ij} = \mu_i + \epsilon_{ij}$$

$$\epsilon_{\{ij\}} \sim N(0, \sigma^2)$$

$$Y_{ij}|\mu_i \overset{ind}{\sim} N(\mu_i, \sigma^2)$$

$$\mu_i = \mu + \tau_i$$

- Random Effects Model

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

$$\tau_i \sim N(0, \sigma_\tau^2), \epsilon_{ij} \sim N(0, \sigma^2)$$

τ_i and ϵ_{ij} are independent

$$Y_{ij}|\tau_i = N(\mu, \sigma\text{^}2) + \tau_i$$

$$Y_{ij} \sim N(\mu, \sigma^2) + N(0, \sigma_\tau{}^2) = N(\mu, \sigma^2 + \sigma_\tau^2)$$

Variance Components:
Because τ_i are random variables, we do not estimate r specific treatment effects. Instead, the model is defined by just two variance parameters: σ_τ^2 (between-group variance) and σ^2 (within-group variance).

Two reference mean definitions

Unweighted mean: treat every group (factor level) as equally important, regardless of how many samples are in each group.
μ is the simple average of the group means:

$$\mu_{\cdot} = \frac{\sum_{i=1}^{r} \mu_i}{r}$$

This replies

$$\sum_{i=1}^{r} \tau_i = 0$$

(The effects are deviations that must cancel each other out across levels).

$$\hat{\mu}_{\cdot} = \frac{\sum_i \bar{Y}_{i\cdot}}{r}, \ \hat{\tau}_i = \bar{Y}_{i\cdot} - \hat{\mu}_{\cdot}$$

Case: When the groups represent distinct categories (e.g., 4 different drug types) and you want a baseline that represents the "average drug," even if one drug trial had 100 people and another only had 20.

Weighted mean: treat every individual data point as equally important. Groups with more samples have a bigger "vote" in determining the grand mean.
μ is the average of the entire population, weighted by group size n_i.

$$\mu_{\cdot} = \sum_{i=1}^{r} w_i \mu_i, \ \text{where} \ \sum_{i=1}^{r} w_i = 1$$

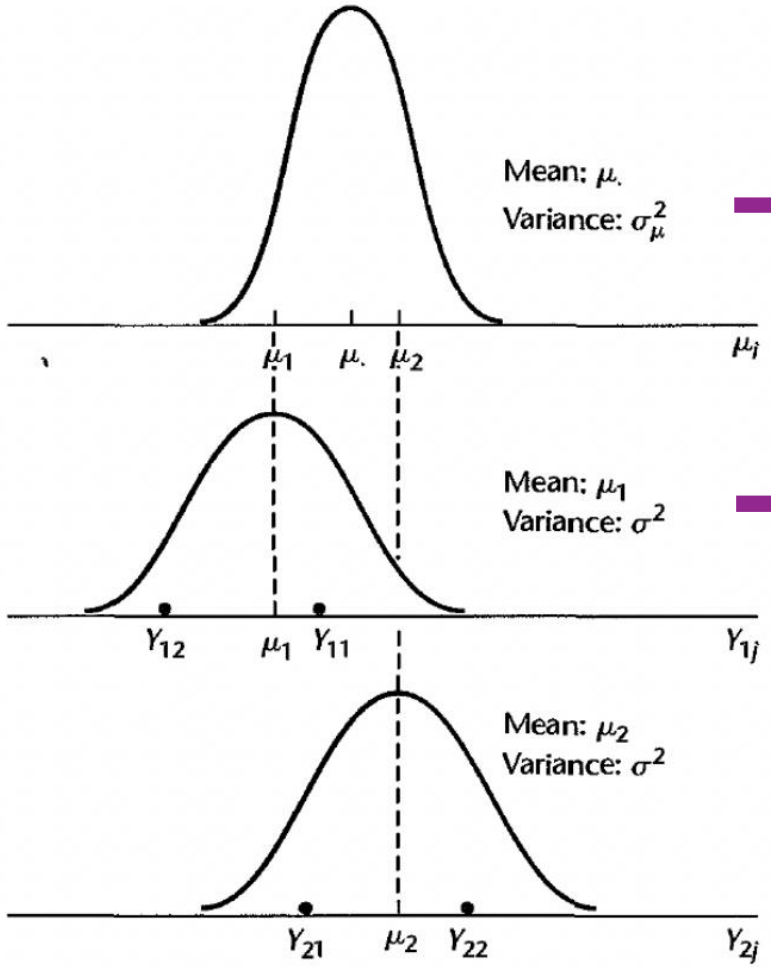
This replies

$$\sum_{i=1}^{r} w_i \tau_i = 0$$

Usually, $w_i = \frac{n_i}{n_T}$ then $\hat{u}_{\cdot} = \bar{Y}_{\cdot\cdot}$ and $\hat{\tau}_i = \bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}$

Case: When the different sample sizes n_i actually reflect the proportions in the real-world population. For example, if you are surveying 15 clinics and the larger clinics represent a larger portion of the patient population.

A perspective from two-stage sampling



Between-Group Variation: population of all possible factor levels

Grand Mean
Between-Group Variance

Within-Group Variation: distribution of individual observations Y_{ij} within the specific groups we sampled.

Group Means
Within-Group Variance

Homoscedasticity: two curves have the exact same shape (width): the error variance is constant across all groups.

The Two-Stage Sampling

1. Sampling Groups: We randomly pick group means from the top population distribution.
2. Sampling Individuals: Once a group mean is "fixed" for that specific study, we sample individual data points Y_{ij} that fluctuate around that mean based on the bottom distribution.

Within-group dependence

Two variance parameters

Cell means are random variables, not parameters

$$Y_{ij} \sim N\left(\mu,\ \sigma_\tau^2 + \sigma^2\right)$$

The observations are not independent: Because observations within the same group (level) share the exact same random effect τ_i , they are no longer independent.

$$\mathrm{Cov}(Y_{ij}, Y_{i'j'}) = 0, \qquad \mathrm{Cov}(Y_{ij}, Y_{ij'}) = \sigma_\tau^2$$

Intraclass correlation coefficient:

$$\mathrm{cor}(Y_{ij}, Y_{ij'}) = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2}$$

ANOVA Table under balanced design

Recall One-Way Fixed Effect

Source	DF	SS	EMS
Trt	$r-1$	$\sum_{i=1}^{r} n(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2$	$\sigma^2+\frac{\sum n(\mu_i-\mu_\cdot)^2}{r-1}$
Error	n_T-r	$\sum\sum(Y_{ij}-\bar{Y}_{i\cdot})^2$	σ^2
Total	n_T-1		

One-Way Random Effect

Source	DF	SS	EMS
Trt	$r-1$	$\sum_{i=1}^{r} n(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2$	$\sigma^2+n\sigma_\mu^2$
Error	n_T-r	$\sum\sum(Y_{ij}-\bar{Y}_{i\cdot})^2$	σ^2
Total	n_T-1		

Estimation in the balanced one-way random-effects model

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \qquad \tau_i \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2), \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

Expected mean squares:

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + n\sigma_\tau^2$$

Variance component estimators:

$$\widehat{\sigma}^2 = MSE$$

$$\widehat{\sigma}_\tau^2 = \frac{MSR - MSE}{n}$$

Inference for the grand mean:

$$\widehat{\mu} = \bar{Y}_{\cdot\cdot}, \qquad \mathrm{Var}(\widehat{\mu}) = \mathrm{Var}(\bar{Y}_{\cdot\cdot}) = \frac{n\sigma_\tau^2 + \sigma^2}{rn}$$

$$\widehat{\mathrm{Var}}(\widehat{\mu}) = \frac{MSR}{rn}$$

$$T = \frac{\bar{Y}_{\cdot\cdot} - \mu}{\sqrt{MSR/(rn)}} \sim t_{r-1}$$

Hypothesis testing under balanced design

In the fixed-effects model, we test equal μ_i : $F_{fixed} = \frac{MSR}{MSE} \sim F_{r-1,nr-r}$, under H_0

In the random effects model under different model assumptions:

$$\frac{\text{MSR}}{\sigma^2 + n\sigma_\mu^2} \sim \frac{\chi^2_{r-1}}{r-1} \qquad \frac{\text{MSE}}{\sigma^2} \sim \frac{\chi^2_{n_T-r}}{n_T-r}$$

MSR is the mean sum of squares due to regression, MSE is the residual mean square

- $H_0: \sigma_\mu^2 = 0$ vs $H_a: \sigma_\mu^2 > 0$
- In the random-effects model, the test statistic is given as

$$\frac{\sigma^2 F_{fixed}}{n\sigma_\mu^2 + \sigma^2} \sim F_{r-1,\, r(n-1)}.$$

Under the hypothesis H: $\sigma_\mu^2 = 0$, the test statistic reduces to F_{fixed}

Summary and Comparison

Feature	Fixed Effects Model	Random Effects Model
Primary Goal	Inference on the specific levels chosen for the experiment.	Inference on the broader population of levels.
Level Selection	Deliberately chosen by the researcher (e.g., 4 specific tomato varieties).	Randomly sampled from a larger population (e.g., 5 random clinics).
Parameters Estimated	Specific group means μ_i (or fixed effects τ_i) and error variance σ^2.	Variance components: between-group variance σ_μ^2 (or σ_τ^2) and error variance σ^2.
Null Hypothesis (H_0)	$H_0: \mu_1 = \mu_2 = \cdots = \mu_r = \mu.$	$H_0: \sigma_\mu^2 = 0.$
Alt. Hypothesis (H_a)	H_a: At least one μ_i is different.	$H_a: \sigma_\mu^2 > 0.$
Scope of Conclusion	Limited strictly to the specific groups tested.	Pertains to the entire population of possible levels.